

BACSWN Executive Summary

BACSWN — Executive Summary (Program Documentation)

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1. Context

BACSWN ("SkyWatch Bahamas") is a sovereign, AI-operated meteorological watch, airspace-surveillance, and emergency-response network for the Commonwealth of The Bahamas, delivered under a **Build-Own-Operate (BOO) public-private partnership** and funded by a share of airline **overflight and landing fees**. The software platform is already built and live; this session produced the **full implementation documentation set** that takes it from a working platform to a buildable, governable national program.

2. What Was Produced This Session

A complete, cross-referenced documentation set (PDF + markdown source), now committed to the project repository under `projects/bacswn/docs/program/`.

#	Document	ID	Audience	Purpose
1	Board Brief (emailed)	—	Board	One-page adoption case (delivered to the board)
2	Board Runbook — Technology Highlights	—	Board	The only board-facing doc; 2-page technology overview
3	Mesh Architecture Master Plan (v1.9, + RF annex)	MARCH-2026-002	Owner/CEO	Controlling engineering blueprint (software + per-tower comms)
4	Master Framework Schedule (v1.8)	SCHED-2026-003	Owner/CEO	24-month, 4-phase, 18-workstream plan with milestone gates
5	Operations & Maintenance Plan	OM-2026-004	Owner/CEO	Day-2 sustainment, NOC, island field logistics, SLAs
6	Data Governance & Sovereignty Policy (v1.1)	GOV-2026-005	Owner/CEO	Data ownership, residency, sharing, trusted AI supply chain
7	Master V&V and ICD Plan	VV-2026-006	Owner/CEO	Test strategy + interface control documents
8	Training & Capacity-Building Plan	TRN-2026-007	Owner/CEO	Bahamian workforce, certification, localization
9	SWOT & Risk Analysis	RISK-2026-008	Owner/CEO	Consolidated living risk register (R01–R20) + SWOT
10	Edge AI Node Hardware Budget	HWB-2026-009	Owner/CEO	Per-tower BOM + hub/NOC room infrastructure

3. Architecture & Design Decisions Captured

- Resilient mesh:** 15 intelligent tower nodes; three-layer comms (licensed VHF mesh + cellular + Iridium satellite); autonomous operation when the hub or an island is cut off.
- Category 4/5 survivability:** structural (TIA-222 / ASCE 7), missile-impact, surge, extended power autonomy, plus lightning/grounding and salt-corrosion protection.
- Full end-to-end cryptography:** hardware root-of-trust, national PKI, signed mesh, FIPS-grade, post-quantum path; plus **hub disaster recovery** (off-island/cloud standby).
- Multimodal edge AI with model consensus:** each tower runs a **multimodal model** (sensors + radar/satellite imagery + text), trained on weather data, in a **2-of-3 model-consensus ensemble** before any autonomous action — distinct from the multi-station consensus. **Models must be non-Chinese / trusted-provenance.**
- Airspace surveillance, three missions, one feed:** commercial **and military** overflight tracking (ADS-B + **Raytheon primary radar**) driving safety, overflight-fee accounting, and CORSIA carbon.
- National-response backbone:** full emergency-services integration (NEMA, Police, Fire/Rescue, RBDF, shelters) via a shared Common Operating Picture and CAP/EDXL.
- Sensing & response extensions:** IoT (LoRaWAN) sensor network and a drone/UAS program (with built-in airspace deconfliction). **Tomorrow.io** space-based radar integrated as a supplementary feed.
- External interoperability (phased):** ICAO OPMET/WAFS and future U.S. (NOAA/NHC, FAA Miami Oceanic, NORAD).

4. Key Findings & Numbers

- **RF reality:** VHF is **horizon-limited, not power-limited**; the remote SE edge (Inagua ↔ Mayaguana, 92 nm) is beyond practical line-of-sight — **satellite is genuinely their primary path** (the link budget validates the design).
 - **Hardware budget** $\sim 1.77M$ * * (*range* * $\sim 1.6M$ – $2.2M$) for 15 nodes + spares + the **hub/NOC room** (racks, UPS, cooling, **fire suppression**, NOC fit-out). **The AI compute is only ~4% of cost** — the budget is dominated by aviation-grade sensors (ceilometer, visibility), resilient power, and the facility.
 - **Program shape:** 24 months, 4 phases, **18 parallel workstreams**, gated milestones; longest long-lead items are URCA spectrum and the **Raytheon radar (ITAR)**.
 - **Risk posture:** 20 tracked risks; the critical few are hurricane-during-deployment, Raytheon/ITAR timing, and cyber/physical attack.
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5. Status

All ten documents are version-controlled and pushed to the gigforge repository ([projects/bacswn/docs/program/](#)). The set is internally consistent and cross-referenced. (*Financial modeling is intentionally out of scope — this is an implementation program, not a financial model.*)

6. Open Decisions & Recommended Next Steps

1. **Model consensus topology** — software 2-of-3 ensemble on one Orin NX (cheaper) **vs.** three independent compute modules (true hardware redundancy, + $\sim$ \$3–4k/node).
 2. **Sensor tiering** — full aviation instrumentation at every site vs. a lighter suite (no ceilometer) at selected remote sites (saves $\sim$ \$300–450k; a meteorological-coverage decision).
 3. **Standby hub** — cloud DR (mostly opex) vs. a physical off-island room.
 4. **Branding / team** — confirm delivery-vehicle attribution; complete the leadership/track-record placeholder in the board runbook.
 5. **Calibration** — review the risk scores and the indicative budget figures against Bahamian context and vendor RFQs.
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Index document for the BACSWN implementation set; see each referenced document for detail.